BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2021

BIOLOGY

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Biology (Objective) Group - I

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** 7465

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. Which animal reproduces by means of budding?

(A) Rhizopus
(B) Hydra
(C) Amoeba
(D) Planarian

کون سا جاندار بڈنگ کے ذریعے ریپروڈکشن کرتا ہے؟

(A) رائزوپس
(B) ہائیڈرا
(C) ایبا
(D) پلینیرین

Q2. The adult human skeleton has bones:

(A) 300
(B) 306
(C) 200
(D) 206

ایک بالغ انسان کے سکیلیٹن میں ہڈیاں ہوتی ہیں:

Q3. Insulin and glucagon are produced in: _____

(A) Adrenal gland
(B) Gonads
(C) Pancreas
(D) Liver

انسولین اور گلوکاگون _____ میں بنتے ہیں۔

(A) ایڈرینل گلینڈ
(B) گونڈز
(C) پینکریاز
(D) جگر

Q4. _____ pairs of spinal nerves arise from spinal cord.

(A) 12
(B) 16
(C) 30
(D) 31

سپائنل کارڈ سے _____ جوڑے سپائنل نروز نکلتے ہیں۔

Q5. Human kidney is _____ centimeter long.

(A) 4
(B) 5
(C) 6
(D) 10

انسانی گردہ _____ سینٹی میٹر لمبا ہوتا ہے۔

Q6. Tobacco smoke contains over _____ different chemicals.

(A) 3000
(B) 4000
(C) 5000
(D) 6000

تمباکو کے دھوئیں میں _____ سے زائد مختلف کیمیکلز ہوتے ہیں۔

Q7. The drug obtained from animals is:

(A) Aspirin
(B) Psilocin
(C) Mescaline
(D) Musk

جانوروں سے حاصل ہونے والی دوا ہے:

(A) اسپرین
(B) سائیلوسن
(C) میسکیلین
(D) مشک

Q8. The complete map of human genome was published in:

(A) 2001
(B) 2002
(C) 2003
(D) 2004

انسانی جینوم کا مکمل نقشہ شائع ہوا:

Q9. Organisms which break down dead complex organic compounds into simple compounds are:

(A) Producers
(B) Consumers
(C) Decomposers
(D) Competitors

آرگنزمز جو مردہ پیچیدہ نامیاتی مرکبات کو سادہ مرکبات میں توڑتے ہیں:

(A) پروڈیوسرز
(B) کنزیومرز
(C) ڈی کمپوزرز
(D) کمپیٹیٹرز

Q10. Which animal is herbivore?

(A) Fox
(B) Frog
(C) Grasshopper
(D) Snake

کون سا جانور ہربی وور ہے؟

(A) لومڑی
(B) مینڈک
(C) ٹڈا
(D) سانپ

Q11. The branch of biology in which we study inheritance is called:

(A) Ecology
(B) Genetics
(C) Pharmacology
(D) Embryology

بائیولوجی کی وہ شاخ جس میں ہم وراثت کا مطالعہ کرتے ہیں کہلاتی ہے:

(A) ایکالوجی
(B) جینیٹکس
(C) فارماکولوجی
(D) ایمبریالوجی

Q12. The phenotypic ratio in Mendel's Law of Independent Assortment was:

(A) 3:1
(B) 1:3
(C) 9:3:3:1
(D) 9:2:2:1

مینڈل کے لا آف انڈیپینڈنٹ اسارٹمنٹ میں فینوٹائپک تناسب تھا:

Biology (Subjective) Group - I

Time Allowed: 01:45 Hours	**Maximum Marks:** 48	**Paper Code:** 7465

Note: Part - I (حصہ اول)

Q2. (Marks 10) Write short answers to any FIVE parts.

(i) Write bad effects of nicotine.

(ii) How will you differentiate between stoma and lenticel?

(iii) Define alveoli.

(iv) Define transpiration.

(v) Write the names of parts of renal tubule.

(vi) Define involuntary actions. Write its one example.

(vii) Write the symptoms of tetany.

(viii) Write the function of thalamus.

Q3. (Marks 10) Write short answers to any FIVE parts.

(i) Differentiate between endoskeleton and exoskeleton.

(ii) Write two examples each of hinge-joints and ball-and-socket joints.

(iii) Define endospore.

(iv) What is alternation of generation?

(v) What are inheritable traits?

(vi) Define allele.

(vii) Identify two traits of pea flower studied by Mendel.

(viii) Write phenotypes of given genotypes in pea: RRYY, rrYY, RRyy, rryy

Q4. (Marks 10) Write short answers to any FIVE parts.

(i) Define species.

(ii) Differentiate between heterotrophs and autotrophs.

(iii) Define parasitism and give an example.

(iv) What is meant by the concept of 3Rs with reference to the conservation of natural resources?

(v) Define genetic engineering.

(vi) Write the uses of formic acid.

(vii) Which drugs are obtained from minerals?

(viii) Which is the most common method of administering vaccine?

> Note: Part - II, Attempt any TWO questions. Each question carries 09 marks.

حصہ دوم، کوئی سے دو سوالات کے جوابات تحریر کیجیے۔ ہر سوال کے 09 نمبر ہیں۔

Q5(a). (Marks 04) Describe the mechanism of removal of extra carbon dioxide and oxygen in plants.

Q5(b). (Marks 05) Define cartilage and explain its types.

Q6(a). (Marks 04) Write a detailed account on pancreas.

Q6(b). (Marks 05) Define fertilization and explain its various types.

Q7(a). (Marks 04) Write a note on drug addiction and associated problems.

Q7(b). (Marks 05) Write basic steps in genetic engineering.

GROUP – II

Biology (Objective) Group - II

Time Allowed: 15 Minutes	**Maximum Marks:** 12	**Paper Code:** 7466

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. There are _____ types of pollination

(A) 2
(B) 4
(C) 3
(D) 5

پولینیشن کی _____ اقسام ہیں۔

Q2. The longest bone of our body is:

(A) Malleus
(B) Incus
(C) Stapes
(D) Thigh

ہمارے جسم کی سب سے لمبی ہڈی ہے:

(A) میلیس
(B) انکس
(C) سٹیپس
(D) ران

Q3. Thyroid gland produces _____ hormone.

(A) Thyroxin
(B) Insulin
(C) Glucagon
(D) Estrogen

تھائی رائیڈ گلینڈ _____ ہارمون پیدا کرتا ہے۔

Q4. Coordinated action has components:

(A) 2
(B) 3
(C) 4
(D) 5

کوآرڈینیٹڈ ایکشن کے اجزاء ہیں:

Q5. Which is a by-product of photosynthesis?

(A) O_2
(B) CO_2
(C) N_2
(D) H_2

کون سی فوٹو سنتھیسز کی بائی پروڈکٹ ہے؟

Q6. Nasal cavity opens into: _____

(A) Pharynx
(B) Oesophagus
(C) Trachea
(D) Lungs

نیزل کیویٹی _____ میں کھلتی ہے۔

(A) فرینکس
(B) ایسوفیگس
(C) ٹریکیا
(D) پھیپھڑے

Q7. Drugs used to reduce pain are known as:

(A) Narcotics
(B) Antiseptics
(C) Antibiotics
(D) Sedatives

درد کم کرنے والی ادویات کہلاتی ہیں:

Q8. Human insulin was prepared in:

(A) 1968
(B) 1978
(C) 1988
(D) 1998

انسانی انسولین _____ میں تیار کی گئی۔

Q9. The term R4 means:

(A) Reduce
(B) Reuse
(C) Recycle
(D) Reforest

اصطلاح R4 کا مطلب ہے:

(A) کم استعمال
(B) بار بار استعمال
(C) ری سائیکلنگ
(D) دوبارہ جنگل لگانا

Q10. There are _____ types of dengue virus.

(A) 2
(B) 3
(C) 4
(D) 5

ڈینگی وائرس کی _____ اقسام ہیں۔

Q11. When was Watson-Crick DNA model presented?

(A) 1943
(B) 1953
(C) 1963
(D) 1973

واٹسن کرک کا ڈی این اے ماڈل کب پیش کیا گیا؟

Q12. A human cell has chromosomes:

(A) 10
(B) 13
(C) 26
(D) 46

ایک انسانی سیل میں کروموسومز کی تعداد ہے:

Biology (Subjective) Group - II

Time Allowed: 01:45 Hours | **Maximum Marks:** 48 | **Paper Code:** 7466

Note: Part - I (حصہ اول)

Q2. (Marks 10) Write short answers to any FIVE parts.

(i) What are lenticels?

(ii) Differentiate between breathing and cellular respiration.

(iii) What is the function of stomata in leaves of plants?

(iv) What is the difference between guttation and dew?

(v) What is glomerulus filtrate?

(vi) What is nerve?

(vii) What is the function of temporal lobe of cerebral cortex?

(viii) Define reflex arc.

Q3. (Marks 10) Write short answers to any FIVE parts.

(i) Differentiate between compact bone and spongy bone.

(ii) Write the name of the longest bone in our body.

(iii) What is multiple fission?

(iv) Write two causes of AIDS.

(v) Define nucleosomes.

(vi) Write the names of nitrogenous bases.

(vii) What is translation?

(viii) Define Mendel's Law of Segregation.

Q4. (Marks 10) Write short answers to any FIVE parts.

(i) Define mutualism.

(ii) Differentiate between biotic and abiotic components of ecosystem.

(iii) Define food web.

(iv) Differentiate between renewable and non-renewable resources.

(v) Name two basic types of fermentation.

(vi) What is meant by recombinant DNA?

(vii) Differentiate between medicinal drug and addictive drug.

(viii) What is the use of tincture of iodine?

> Note: Part - II, Attempt any TWO questions. Each question carries 09 marks.

> حصہ دوم، کوئی سے دو سوالات کے جوابات تحریر کیجیے۔ ہر سوال کے 09 نمبر ہیں۔

Q5(a). (Marks 04) Explain osmoregulatory function of kidney.

Q5(b). (Marks 05) Explain the components of human skeleton.

Q6(a). (Marks 04) Write a note on neuron.

Q6(b). (Marks 05) Explain the methods of artificial vegetative propagation in plants.

Q7(a). (Marks 04) Describe the mode of action of vaccine.

Q7(b). (Marks 05) Describe any five achievements of genetic engineering.